$$\begin{cases} \dfrac{\partial \ell}{\partial a_1} = 0 \\[2ex] \dfrac{\partial \ell}{\partial a_2} = 0 \end{cases}$$

$$\begin{cases} \sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i = 0 & (1) \\[2ex] \sum_{i=1}^{75} x_i y_i - a_1 \sum_{i=1}^{75} x_i - a_2 \sum_{i=1}^{75} x_i^2 = 0 & (2) \end{cases}$$

Solve for a_1 in eq. (1).

$$75a_1 = \sum_{i=1}^{75} y_i - a_2 \sum_{i=1}^{75} x_i$$

$$a_1 = \frac{1}{75}\left(\sum_{i=1}^{75} y_i - a_2 \sum_{i=1}^{75} x_i\right) = \frac{1}{75}\sum_{i=1}^{75} y_i - \left(\frac{1}{75}\sum_{i=1}^{75} x_i\right) a_2$$

Substitute a_1 into eq. (2) & solve for a_2.

$$\sum_{i=1}^{75} x_i y_i - \sum_{i=1}^{75} x_i \left(\frac{1}{75}\sum_{i=1}^{75} y_i - \left(\frac{1}{75}\sum_{i=1}^{75} x_i\right) a_2\right) - \left(\sum_{i=1}^{75} x_i^2\right) a_2 = 0$$

$$\sum_{i=1}^{75} x_i y_i - \frac{1}{75}\sum_{i=1}^{75} x_i \sum_{i=1}^{75} y_i = \left(\sum_{i=1}^{75} x_i^2\right) a_2 - \frac{1}{75}\left(\sum_{i=1}^{75} x_i\right)^2 a_2$$

$$\sum_{i=1}^{75} x_i y_i - \frac{1}{75}\sum_{i=1}^{75} x_i \sum_{i=1}^{75} y_i = \left[\sum_{i=1}^{75} x_i^2 - \frac{1}{75}\left(\sum_{i=1}^{75} x_i\right)^2\right] a_2$$

$$a_2 = \frac{\sum_{i=1}^{75} x_i y_i - \frac{1}{75}\sum_{i=1}^{75} x_i \sum_{i=1}^{75} y_i}{\sum_{i=1}^{75} x_i^2 - \frac{1}{75}\left(\sum_{i=1}^{75} x_i\right)^2}$$

$$\frac{\partial \ell}{\partial a_2} = \frac{\partial}{\partial a_2}\left(75 \ln\left(\frac{1}{\sqrt{2\pi}}\right) + \sum_{i=1}^{75} -\frac{1}{2}(y_i - a_1 - a_2 x_i)^2\right)$$

$$= 0 + \sum_{i=1}^{75} \frac{\partial}{\partial a_2}\left(-\frac{1}{2}(y_i - a_1 - a_2 x_i)^2\right)$$

$$= \sum_{i=1}^{75} -\frac{1}{2}\left(\frac{\partial}{\partial a_2}(y_i - a_1 - a_2 x_i)^2\right)$$

$$= \sum_{i=1}^{75} -\frac{1}{2}\left(2(y_i - a_1 - a_2 x_i)(-x_i)\right)$$

$$= \sum_{i=1}^{75} y_i x_i - a_1 x_i - a_2 x_i^2$$

$$= \sum_{i=1}^{75} y_i x_i + \sum_{i=1}^{75} -a_1 x_i + \sum_{i=1}^{75} -a_2 x_i^2$$

$$= \sum_{i=1}^{75} x_i y_i + (-a_1)\sum_{i=1}^{75} x_i + (-a_2)\sum_{i=1}^{75} x_i^2$$

$$= \sum_{i=1}^{75} x_i y_i - a_1 \sum_{i=1}^{75} x_i - a_2 \sum_{i=1}^{75} x_i^2$$

$$\ell(a_1, a_2) = \ln\left(\prod_{i=1}^{75} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y_i - a_1 - a_2 x_i)^2}\right)$$

$$= \sum_{i=1}^{75} \ln\left(\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y_i - a_1 - a_2 x_i)^2}\right)$$

$$= \sum_{i=1}^{75} \left[\ln\left(\frac{1}{\sqrt{2\pi}}\right) + \ln e^{-\frac{1}{2}(y_i - a_1 - a_2 x_i)^2}\right]$$

$$= \sum_{i=1}^{75} \left[\ln\left(\frac{1}{\sqrt{2\pi}}\right) + \left(-\frac{1}{2}\right)(y_i - a_1 - a_2 x_i)^2\right]$$

$$= \sum_{i=1}^{75} \ln\left(\frac{1}{\sqrt{2\pi}}\right) + \sum_{i=1}^{75} -\frac{1}{2}(y_i - a_1 - a_2 x_i)^2$$

$$= \ln\left(\frac{1}{\sqrt{2\pi}}\right) \cdot \sum_{i=1}^{75} 1 + \sum_{i=1}^{75} -\frac{1}{2}(y_i - a_1 - a_2 x_i)^2$$

$$= \ln\left(\frac{1}{\sqrt{2\pi}}\right)(75) + \sum_{i=1}^{75} -\frac{1}{2}(y_i - a_1 - a_2 x_i)^2$$

$$= 75 \ln\left(\frac{1}{\sqrt{2\pi}}\right) + \sum_{i=1}^{75} -\frac{1}{2}(y_i - a_1 - a_2 x_i)^2$$

$$\frac{\partial \ell}{\partial a_1} = \frac{\partial}{\partial a_1}\left(75 \ln\left(\frac{1}{\sqrt{2\pi}}\right) + \sum_{i=1}^{75} -\frac{1}{2}\left(y_i - a_1 - a_2 x_i^2\right)\right)$$

$$= 0 + \sum_{i=1}^{75} \frac{\partial}{\partial a_1}\left[-\frac{1}{2}(y_i - a_1 - a_2 x_i)^2\right]$$

$$= \sum_{i=1}^{75} -\frac{1}{2}\left(\frac{\partial}{\partial a_1}(y_i - a_1 - a_2 x_i)^2\right)$$

$$= \sum_{i=1}^{75} -\frac{1}{2}\left(2(y_i - a_1 - a_2 x_i)(-1)\right)$$

$$= \sum_{i=1}^{75} y_i - a_1 - a_2 x_i$$

$$= \sum_{i=1}^{75} y_i + \sum_{i=1}^{75} a_1 + \sum_{i=1}^{75} -a_2 x_i$$

$$= \sum_{i=1}^{75} y_i + (-a_1)\sum_{i=1}^{75} 1 + (-a_2)\sum_{i=1}^{75} x_i$$

$$= \sum_{i=1}^{75} y_i - a_1(75) - a_2 \sum_{i=1}^{75} x_i$$

$$= \sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i$$

$$\frac{\partial^2 \ell}{\partial a_1^2} = \frac{\partial}{\partial a_1}\left(\frac{\partial \ell}{\partial a_1}\right)$$

$$= \frac{\partial}{\partial a_1}\left(\sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i\right)$$

$$= 0 - 75 - 0$$

$$= -75$$

$$\frac{\partial^2 \ell}{\partial a_2^2} = \frac{\partial}{\partial a_2}\left(\frac{\partial \ell}{\partial a_2}\right)$$

$$= \frac{\partial}{\partial a_2}\left(\sum_{i=1}^{75} x_i y_i - a_1 \sum_{i=1}^{75} x_i - a_2 \sum_{i=1}^{75} x_i^2\right)$$

$$= 0 - 0 - \sum_{i=1}^{75} x_i^2$$

$$= -\sum_{i=1}^{75} x_i^2$$

$$\frac{\partial^2 \ell}{\partial a_1 \partial a_2} = \frac{\partial}{\partial a_2}\left(\frac{\partial \ell}{\partial a_1}\right)$$

$$= \frac{\partial}{\partial a_2}\left(\sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i\right)$$

$$= 0 - 0 - \sum_{i=1}^{75} x_i$$

$$= -\sum_{i=1}^{75} x_i$$

Note: $\frac{\partial^2 \ell}{\partial a_1 \partial a_2} = \frac{\partial^2 \ell}{\partial a_2 \partial a_1}$ Since $\ell(a_1, a_2)$ is a nice function.

$$\frac{\partial^2 \ell}{\partial a_1^2} = \frac{\partial}{\partial a_1}\left(\frac{\partial \ell}{\partial a_1}\right)$$

$$= \frac{\partial}{\partial a_1}\left(\sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i\right)$$

$$= 0 - 75 - 0$$

$$= -75$$

$$\frac{\partial^2 \ell}{\partial a_2^2} = \frac{\partial}{\partial a_2}\left(\frac{\partial \ell}{\partial a_2}\right)$$

$$= \frac{\partial}{\partial a_2}\left(\sum_{i=1}^{75} x_i y_i - a_1 \sum_{i=1}^{75} x_i - a_2 \sum_{i=1}^{75} x_i^2\right)$$

$$= 0 - 0 - \sum_{i=1}^{75} x_i^2$$

$$= -\sum_{i=1}^{75} x_i^2$$

$$\frac{\partial^2 \ell}{\partial a_1 \partial a_2} = \frac{\partial}{\partial a_2}\left(\frac{\partial \ell}{\partial a_1}\right)$$

$$= \frac{\partial}{\partial a_2}\left(\sum_{i=1}^{75} y_i - 75a_1 - a_2 \sum_{i=1}^{75} x_i\right)$$

$$= 0 - 0 - \sum_{i=1}^{75} x_i$$

$$= -\sum_{i=1}^{75} x_i$$